

Sri Lanka AI Challenge 2025 (SLAIC 2025): Case Study

Smart Transit Companion: AI-Driven Multi-Modal Mobility Assistant

Background

Public transport is the backbone of mobility for millions of Sri Lankans, spanning buses, trains, tuk-tuks, and other informal modes. However, the ecosystem is fragmented, analog, and unpredictable, with limited real-time data, poor integration, and minimal user-centric innovation.

With smartphone adoption growing across all income levels, there is an opportunity to introduce an AI-powered, agent-driven transit companion that delivers personalized, real-time, multi-modal travel assistance, improving accessibility, reliability, and commuter experience.

Problem Context

Commuters in Sri Lanka face multiple systemic challenges:

1. **Fragmented and Outdated Information:** No unified, authoritative source for accurate bus/train schedules or real-time updates.
2. **Lack of Centralized Digital Access:** No single platform for multi-modal route planning, fare comparisons, and live updates.
3. **Limited Inclusivity:** Language barriers and inaccessible interfaces for differently abled users.
4. **No Personalized or Context-Aware Guidance:** Routes and timing are not optimized for individual preferences, constraints, or travel purposes.
5. **Poor Disruption Visibility:** No real-time alerts for delays, strikes, or diversions.
6. **Weak Multi-Modal Integration:** No seamless handoff between buses, trains, and last-mile options.
7. **Tourist & Non-Local Challenges:** Non-residents struggle without digital navigation aids.

8. **No Natural Language Query Support:** Commuters cannot easily ask travel-related questions in plain language.
9. **Rural & Suburban Data Gaps:** Limited digitization for outstation and rural transport services.

Agentic AI Opportunity

The Smart Transit Companion will operate as a multi-agent AI ecosystem, each agent with a distinct responsibility:

- **Data Aggregation Agent:** Collects and unifies real-time transport data from APIs, GTFS feeds, scraping, and crowdsourced inputs.
- **Route Optimization Agent:** Designs multimodal travel plans that minimize travel time, wait time, and cost.
- **Disruption Management Agent:** Detects delays or service interruptions and dynamically reroutes users.
- **Personalization Agent:** Learns user preferences for travel modes, timings, and accessibility needs.
- **Language & Accessibility Agent:** Offers multilingual, voice-assisted, and visually adaptive interfaces.
- **Fare Optimization Agent:** Identifies the lowest-cost travel combinations, including passes, discounts, and offers.
- **Local Knowledge Agent:** Supplements official data with community-contributed updates, especially for rural and semi-urban areas.

Expected User Outcomes

- **50% Reduction in commuter uncertainty** through real-time guidance and disruption alerts.
- **Seamless multi-modal integration** enabling faster, more efficient travel.
- **Inclusive access** for non-locals, tourists, and differently abled passengers.
- **Increased public transport adoption** due to improved reliability and transparency.

Data & API Resources

Participants may use real, mock, or hybrid datasets. Suggested sources include (but are not limited to):

- [Sri Lanka Railways Location API](#) – Real-time train GPS data.
- [Sri Lanka Open Data Portal](#) – Transport-related datasets.
- [GTFS Standard](#) – For structuring transport schedules.
- [Inter-Provincial Bus Timetables](#) – Timetables regulated by NTC.
- [Inter-Provincial Bus Route Maps](#) – Route maps regulated by NTC.
- [Inter-Provincial Bus Fares](#) – Fare structures regulated by NTC.

Structured Outcome

The Smart Transit Companion brings together agentic AI, real-time data, and user personalization to transform how Sri Lankans experience public transport. By bridging the gap between fragmented services, it has the potential to make commuting smarter, faster, and more inclusive, while contributing to sustainable urban and rural mobility.

Approach

The Smart Transit Companion will transform fragmented and outdated public transport information into a unified, real-time, and user-tailored travel experience.

It will begin by aggregating schedule, route, and disruption data from multiple sources, such as official APIs, GTFS feeds, operator websites, crowdsourced updates, and historical travel patterns, into a central knowledge layer.

Using this foundation, specialized AI agents will generate optimized, multi-modal travel plans based on the user's departure point, destination, preferred modes, budget, language, and accessibility requirements. The system will continuously monitor for service changes or delays, proactively adjusting routes and notifying users of better alternatives.

By combining official operator data with community-sourced intelligence, the companion will extend coverage to rural and under-digitized areas, while multilingual and accessible interfaces ensure inclusivity.

The end goal is to deliver faster, more reliable, and more transparent journeys, whether for a daily commuter, a tourist navigating a new city, or a rural passenger connecting to essential services.